

CHE 201 Fall 2019 HW 11

Alex Boskov

TOTAL POINTS

95 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **5 pts** u_1, u_2 incorrect, so Q also incorrect
- **5 pts** S_1, S_2 incorrect
- **1 pts** ΔS unit incorrect
- **3 pts** ΔS answer missing
- **1 pts** ΔQ is negative in this case
- **8 pts** ΔS equation for idea gas was incorrect

QUESTION 2

2 2 15 / 15

✓ - **0 pts** Correct

- **2 pts** ΔS is positive

QUESTION 3

3 3 15 / 15

✓ - **0 pts** Correct

- **7.5 pts** $W = ?$
- **2 pts** Work is negative
- **2 pts** Q is negative
- **3.5 pts** Wrong value for s_1 . $s_1 = 1.9690 \text{ Btu}/(\text{lb} \cdot \text{R})$
- **2 pts** Wrong conversion from $^{\circ}\text{F}$ to $^{\circ}\text{R}$

QUESTION 4

4 4 15 / 15

✓ - **0 pts** Correct

- **5 pts** $T_h = 336 \text{ K}$ (63°C)
- **5 pts** $\text{CoP} = T_c/(T_h - T_c)$
- **15 pts** No answer

QUESTION 5

5 5 20 / 20

✓ - **0 pts** Correct

- **2 pts** c) the rate of entropy production is positive
- **3 pts** a) $W_{\text{electric}} = 2,990 \text{ kW}$. $Q = -158 \text{ kW}$.

- **6 pts** c) rate of entropy production = $-Q/T_o = 0.53 \text{ kW/K}$

QUESTION 6

6 6 15 / 20

- **0 pts** correct

- **5 pts** (1) definition of ΔS_p ; (2) why $\Delta S_p = 0$; (3) why $s_{1f}, s_{1i}, s_{2f}, s_{2i}$ were not included for the following calculation?

✓ - **3 pts** Its **400K** based on the given question

- **1 pts** unit of K not degree K

- **5 pts** The ΔS equation for idea gas was incorrect

- **6 pts** final $P = ?$

✓ - **2 pts** entropy produce calculation incorrect, $P = 400 \text{ not } 300 \text{ K}$

- **6 pts** entropy produce = ?

- **20 pts** no submission

- **5 pts** ΔS calculation incorrect

- **1 pts** Unit for T_2

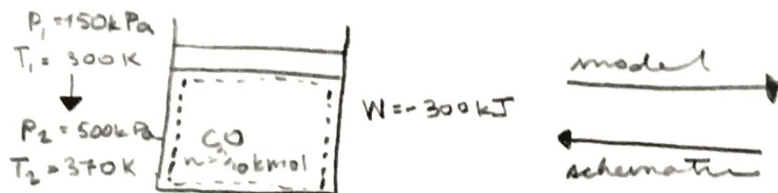
- **3 pts** calculation of m_1 & $m_2 = ?$ Final P incorrect

- **3 pts** (1) why the delta piston was mentioned at the beginning and was not included in the following calculation? need the explanation for each step if it was neglected

12/07/2019

BOSKOV ALEX (1)

1



- ideal gas
- closed system
- $\Delta KE, \Delta PE$ negligible
- work is expansion/compression

$$\cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = Q - W$$

$$\Delta U = Q - W$$

$$(a) \Delta U = n(u_2 - u_1)$$

$$u_1 \xrightarrow{A-23} 6229 \frac{\text{kJ}}{\text{kmol}}$$

$$u_2 \xrightarrow{A-23} 7689 \frac{\text{kJ}}{\text{kmol}}$$

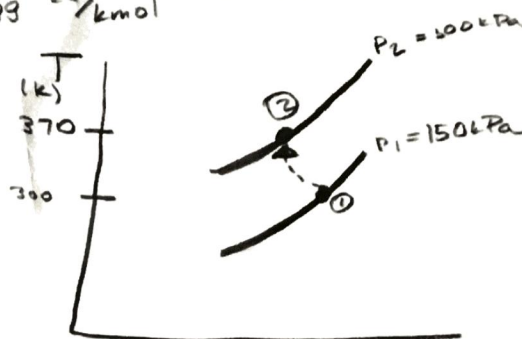
$$n = \frac{1}{10} \text{ kmol}$$

$$= \frac{1}{10} (7689 - 6229) = 146 \text{ kJ}$$

$$\Delta U = Q - W$$

$$(146 \text{ kJ}) = Q - (-300 \text{ kJ})$$

$$\boxed{Q = -154 \text{ kJ}} \text{ from the system}$$



$$(b) \Delta S = S^\circ(T_2) - S^\circ(T_1) - R \ln \frac{P_2}{P_1}$$

$$\Delta S = (203.842 - 197.723) - 8.314 \ln \left(\frac{500}{150} \right)$$

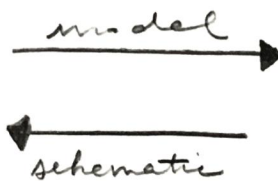
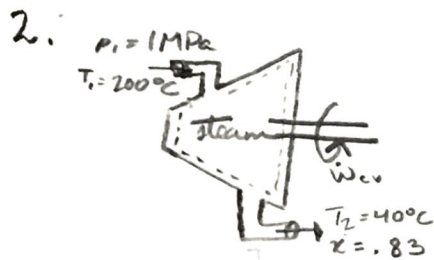
$$\Delta S = -3.89 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$S^\circ(T_1) \xrightarrow{A-23} 197.723 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$S^\circ(T_2) \xrightarrow{A-23} 203.842 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$R = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$\rightarrow -3.89 \cdot \frac{1}{10} = \boxed{-0.389 \frac{\text{kJ}}{\text{K}}}$$



- control volume
- 1 inlet 1 outlet
- steady-state
- $\Delta KE = \Delta PE = \dot{Q}_{cv} = 0$
- work in form of shaft work

$$(a) \dot{m}_i = \dot{m}_e = \dot{m}$$

$$\text{energy balance} \left[\cancel{\dot{Q}_{cv}} = \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m} [(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}} + g(\cancel{z_1 - z_2})] \right]$$

$$\dot{W}_{cv} = \dot{m} (h_1 - h_2)$$

$$\frac{\dot{W}_{cv}}{\dot{m}} = h_1 - h_2$$

$$h_1 \xrightarrow{A-4} 2827.9 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 \xrightarrow{A-2} (0.83)(2574.3) + (0.17)(167.57) = 2165.2 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{\dot{W}_{cv}}{\dot{m}} = (2827.9 - 2165.2) = \boxed{662.7 \frac{\text{kJ}}{\text{kg}}}$$

$$(b) s_1 \xrightarrow{A-4} 6.694 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$s_2 \xrightarrow{A-2} (0.83)(5.725) + (0.17)(8.1648) = 6.95 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$\Delta S = 6.95 - 6.694 = \boxed{0.257 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} = \Delta S}$$

11 15 / 15

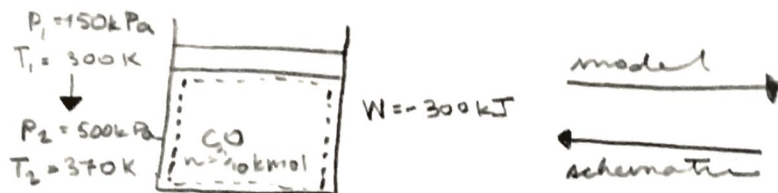
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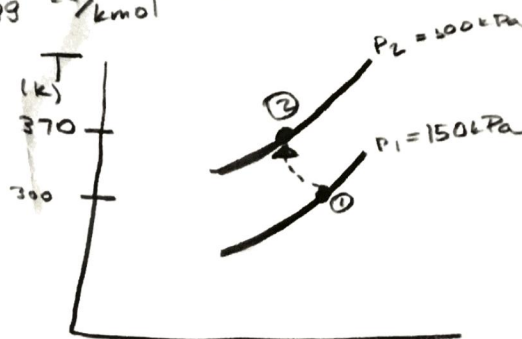
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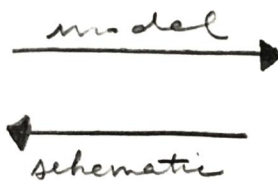
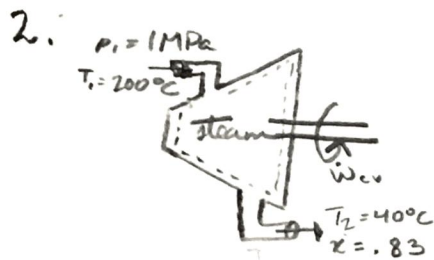
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2 2 15 / 15

✓ - 0 pts Correct

- 2 pts Delta S is positive

3

$$A = 10 \text{ psi}$$

$$T_1 = 500^\circ \text{F}$$

$$P_2 = 80 \text{ psi}$$

$$T_2 = 800^\circ \text{F}$$



model
schematic

- internally reversible process ($\sigma = 0$)
- closed system
- Temperature $\sim$ specific entropy
- $\Delta KE = \Delta PE = 0$

$$s_2 - s_1 = \int_1^2 \left(\frac{\delta Q}{T} \right) + \cancel{\sigma} \rightarrow \Delta s \cdot T = Q_{12}$$

$$s_1 \xrightarrow{A-4} 1.9690 \text{ Btu/lb} \cdot ^\circ \text{R}$$

$$s_2 \xrightarrow{A-4} 1.8700 \text{ Btu/lb} \cdot ^\circ \text{R}$$

$$T_1 = 500 + 459.67 = 959.67^\circ \text{R}$$

$$T_2 = 800 + 459.67 = 1259.67^\circ \text{R}$$

$$T_{\text{avg}} = \frac{959.67 + 1259.67}{2} = 1109.67^\circ \text{R}$$

$$-(1.9690 - 1.8700)(1109.67) = Q_{12} = \boxed{-109.9 \text{ Btu/lb}} \text{ from system: heat}$$

$$\cancel{\Delta KE} + \cancel{\Delta KE} + \Delta U = Q - W$$

$$W = Q - \Delta U$$

$$u_1 \xrightarrow{A-4} 1182.2 \text{ Btu/lb}$$

$$u_2 \xrightarrow{A-4} 1292.4 \text{ Btu/lb}$$

$$W = (-109.9) - (1292.4 - 1182.2) = \boxed{-220.1 \text{ Btu/lb}} \text{ on system: work}$$

4.

$$s_2 \xrightarrow{A-10} 0.9190 \text{ kJ/kg} \cdot \text{K} \text{ at } T_1 = 0^\circ \text{C}$$

interpolate to find T_2

$$\frac{0.9457 - 0.9069}{10 - 60} = \frac{0.9190 - 0.9069}{T_2 - 60} \text{ at } T_2 = 63.11^\circ \text{C}$$

COP of
Carnot refrigerator
cycle

$$\frac{T_c}{T_H - T_c} = \frac{(0 + 273) \text{ K}}{(63.11 + 273) \text{ K} - (0 + 273) \text{ K}} = \boxed{4.32}$$

33 15 / 15

✓ - 0 pts Correct

- 7.5 pts $W = ?$

- 2 pts Work is negative

- 2 pts Q is negative

- 3.5 pts Wrong value for $s1$. $s1 = 1.9690 \text{ Btu}/(\text{lb}\cdot\text{R})$

- 2 pts Wrong conversion from oF to oR

3

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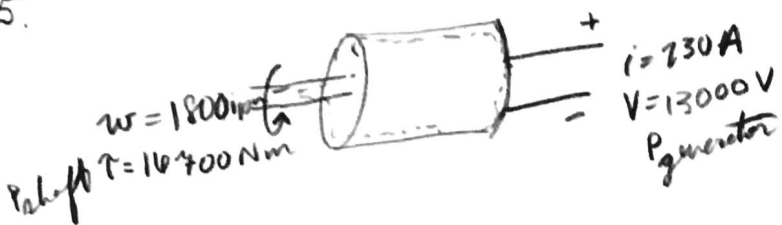
4 4 15 / 15

✓ - 0 pts Correct

- 5 pts $T_h = 336 \text{ K}$ (63 °C)

- 5 pts $\text{CoP} = T_c / (T_h - T_c)$

- 15 pts No answer



model
schematic

BOSKOV ALEX (3)
steady state
shaft work
electrical work

$$(a) \dot{Q} = -P_{\text{shaft}} + P_{\text{generator}}$$

$$P_{\text{shaft}} \rightarrow \frac{1800 \text{ rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ sec}} = 188.5 \frac{\text{rad}}{\text{s}}$$

$$(16700)(188.5) = 3147875.8 \text{ W}$$

$$P_{\text{generator}} \rightarrow P = IV = (230)(13000) = 2990000 \text{ W}$$

$$\dot{Q} = -(3147875.8) + 2990000 = -157875.8 \text{ W}$$

$$\dot{Q} = -hA(T_b - T_o) \quad h = 110 \text{ W/m}^2 \cdot \text{K} \quad T_o = 298 \text{ K} \quad A = 32 \text{ m}^2$$

$$-157875.8 = -(110)(32)(T_b - 298)$$

$$\boxed{T_b = 342.9 \text{ K}}$$

$$(b) \cancel{S_{\text{gen}}} = \int \left(\frac{-\delta Q}{T_b} \right) + \sigma$$

$$\dot{\sigma} = \frac{\dot{Q}}{T_b} = \frac{(110)(32)(342.9 - 298)}{342.9 \times 1000}$$

$$= \boxed{0.4604 \text{ kW/K}}$$

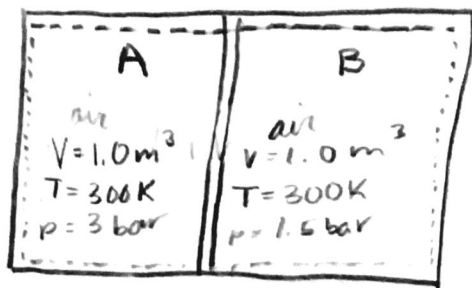
$$(c) \sigma = \frac{\dot{Q}}{T_o} = \frac{(110)(32)(342.9 - 298)}{298 \times 1000} \quad \text{convert to kW}$$

$$= \boxed{0.5297 \text{ kW/K}}$$

5 5 20 / 20

✓ - 0 pts Correct

- 2 pts c) the rate of entropy production is positive
- 3 pts a) $W_{\text{electric}} = 2,990 \text{ kW}$. $Q = -158 \text{ kW}$.
- 6 pts c) rate of entropy production = $-Q/T_o = 0.53 \text{ kW/K}$



BOSKOV ALEX (4) ($R_{air} = 0.287 \frac{kJ}{kg \cdot K}$)

- ideal gas
- model for an insulated chamber ($Q=0$)
- no state change for piston ($W=0$)

$$(a) \quad m_A = \frac{PV}{RT} = \frac{(3 \text{ bar}) \left(\frac{10^2 \text{ kN/m}^2}{1 \text{ bar}} \right) (1)}{(0.287)(300)} = 3.48 \text{ kg}$$

$$m_B = \frac{PV}{RT} = \frac{(1.5 \text{ bar}) \left(\frac{10^2 \text{ kN/m}^2}{1 \text{ bar}} \right) (1)}{(0.287)(300)} = 1.742 \text{ kg}$$

$$\Delta U = \cancel{X} - \cancel{X}$$

$$\Delta U = 0$$

$$m_A(T_2 - 300) = m_B(T_2 - 300) = 0$$

$$\boxed{T_2 = 300 \text{ K}}$$

$$(b) \quad P_{2A} = P_{2B}$$

$$\frac{m_A R T_2}{V_{2A}} = \frac{m_B R T_2}{V_{2B}} \rightarrow \frac{m_A}{m_B} = \frac{V_{2A}}{V_{2B}}$$

$$V_{2A} = 2V_{2B}$$

$$\left(\frac{2}{3}\right)(2) = 1.33 \text{ m}^3 = V_{2A} \quad \left(\frac{1}{3}\right)(2) = 0.667 \text{ m}^3 = V_{2B}$$

$$P = \frac{(3.48)(R)(300)}{1.33 \text{ m}^3} = (225 \text{ kPa}) \left(\frac{1 \text{ bar}}{10^2 \text{ kPa}} \right) = \boxed{2.25 \text{ bar}}$$

$$(c) \quad \Delta S_A = m_A c_v \ln\left(\frac{T_{2A}}{T_{1A}}\right) + m_A R \ln\left(\frac{V_{2A}}{V_{1A}}\right)$$

$$= (3.48)(.718)(0) + (3.48)(0.287) \ln\left(\frac{1.33}{1.0}\right)$$

$$= 0.285 \text{ kJ/K}$$

$$\Delta S_B = m_B c_v \ln\left(\frac{T_{2B}}{T_{1B}}\right) + m_B R \ln\left(\frac{V_{2B}}{V_{1B}}\right)$$

$$\Delta S_B = -0.203 \text{ kJ/K}$$

$$\Delta S_{\text{total}} = 0.285 - 0.203 = \boxed{0.0821 \text{ kJ/K}}$$

66 15 / 20

- 0 pts correct

- 5 pts (1) definition of ΔS_p ; (2) why $\Delta S_p = 0$; (3) why s_{1f} , s_{1i} , s_{2f} , s_{2i} were not included for the following calculation?

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